OS ORANGE ASSIGNMENT

SRN: PES2UG24CS087

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Github link: https://github.com/mgarya/PES2UG24CS087-pes-vcs.git

Screenshots:

Phase 1

1 A) Output of `./test_objects`

```
mg_arya@LAPTOP-8O35GT53:~$ cd /mnt/c/OS-Orange/PES2UG24CS087-pes-vcs
make clean
make test_objects
./test_objects
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1 B) Output of `find .pes/objects -type f`

```
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$
```

Phase 2

2 A) Output of `./test_tree`

```
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$
```

2 B) Raw binary of tree object

```
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ xxd .pes/objects/2c/f8d83d9ee29543b34a877274
21fdecb7e3f3a183d337639025de576db9ebb4
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
```

Phase 3

3 A) pes init -> pes add -> pes status

```
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ echo "hello" > file1.txt
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ echo "world" > file2.txt
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ ./pes add file1.txt file2.txt
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h
```

3 B) cat .pes/index output

```
mg_arya@LAPTOP-8035GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776442696 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776442711 6 file2.txt
```

Phase 4

4 A) pes log output

```
./pes log
Initialized empty PES repository in .pes/
Committed: 31725d354146... Initial commit
Committed: bb655589a39d... Add world
Committed: 35cf14b0fe92... Add farewell
commit 35cf14b0fe920f7281b8e15625545bb1c82562de80bfab788347f26f92ed8db0
Author: M G Arya <PES2UG24CS087>
Date:   1776443133

    Add farewell

commit bb655589a39d33c81603a121b08416d5dc588d79d8572fad2072de81cbcccead
Author: M G Arya <PES2UG24CS087>
Date:   1776443133

    Add worldell

commit 31725d354146c5b3f501fb880c158dbd6185c2ce6ad66ab609763f37a9e28ec3
Author: M G Arya <PES2UG24CS087>
Date:   1776443132

    Initial commit
```

4 B) `find .pes -type f | sort` showing object growth

```
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/31/725d354146c5b3f501fb880c158dbd6185c2ce6ad66ab609763f37a9e28ec3
.pes/objects/35/cf14b0fe920f7281b8e15625545bb1c82562de80bfab788347f26f92ed8db0
.pes/objects/55/8c1061e21bccbeb23c527d8dcef5c8646cabfb6091aae3ae7ee3320c9ff7b7
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/bb/655589a39d33c81603a121b08416d5dc588d79d8572fad2072de81cbcccead
.pes/objects/cc/e49eb2943f54e8135f291862bdf218064409ba05a7ee7d5bec00ddc22addfb
.pes/objects/dd/82340b5e403e2dc18bfca050e6699935849710b2b5d28ab8bbffbb72d32627
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4 C) cat .pes/refs/heads/main and cat .pes/HEAD

```
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ cat .pes/refs/heads/main
35cf14b0fe920f7281b8e15625545bb1c82562de80bfab788347f26f92ed8db0
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Final Full integration test

```
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 31f9f688e0a0... Initial commit

Log after first commit:
commit 31f9f688e0a01810014f22f8304046d1cbf17befcc414bbf6f829b23b65fcd92
Author: M G Arya <PES2UG24CS087>
Date:   1776443360

    Initial commit
```

```
--- Second Commit ---
Committed: 6e8a9b7ef677... Update file.txt

--- Third Commit ---
Committed: 119e7d96266d... Add farewell

--- Full History ---
commit 119e7d96266d0fab7ddcdae42a1d45a6d62530c6ca43454ed97df39ec269dd06
Author: M G Arya <PES2UG24CS087>
Date:   1776443360

    Add farewell

commit 6e8a9b7ef677a0726a5a1a50fd7629c918670b6b88d4fc8e4f1cd0610e37e09a
Author: M G Arya <PES2UG24CS087>
Date:   1776443360

    Update file.txt

commit 31f9f688e0a01810014f22f8304046d1cbf17befcc414bbf6f829b23b65fcd92
Author: M G Arya <PES2UG24CS087>
Date:   1776443360

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
119e7d96266d0fab7ddcdae42a1d45a6d62530c6ca43454ed97df39ec269dd06

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/11/9e7d96266d0fab7ddcdae42a1d45a6d62530c6ca43454ed97df39ec269dd06
.pes/objects/31/f9f688e0a01810014f22f8304046d1cbf17befcc414bbf6f829b23b65fcd92
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/8a9b7ef677a0726a5a1a50fd7629c918670b6b88d4fc8e4f1cd0610e37e09a
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
```

```
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/11/9e7d96266d0fab7ddcdae42a1d45a6d62530c6ca43454ed97df39ec269dd06
.pes/objects/31/f9f688e0a01810014f22f8304046d1cbf17befcc414bbf6f829b23b65fcd92
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/8a9b7ef677a0726a5a1a50fd7629c918670b6b88d4fc8e4f1cd0610e37e09a
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
mg_arya@LAPTOP-8O35GT53:/mnt/c/OS-Orange/PES2UG24CS087-pes-vcs$
```

Phase 5 and 6 Analysis question answers

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename. Resolve the target snapshot by reading the commit object, its root tree, and recursively expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.

This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash. Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state.

If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse.

This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

In detached HEAD, .pes/HEAD stores a commit hash directly instead of a branch ref, and new commits still work by pointing to the previous detached commit while HEAD moves forward. The risk is that no branch references these commits, making them potentially unreachable.

Recovery involves creating a branch ref pointing to that commit chain or switching to a branch after anchoring the detached commit. If not recovered and no refs point to them, future GC may delete them.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while

enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories.

A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects.

Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.